Band-1 Hadamard order 14304 – Classical Paley-I (q=14303); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14304 Paley-I PASS; Paley-II FAIL/SKIP (q=7151 prime but ===3 mod 4 needs ===1); Kronecker FAIL/SKIP (4*3576; 3576===0 mod 4 order-admissible but H_3576 has NO direct classical Paley route); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-09

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-09_Band1_Hadamard_14252_PaleyI`; `2026-10-09_Band1_Hadamard_14244_PaleyI`; `2026-10-09_Band1_Hadamard_14208_PaleyI`; `2026-10-08_Band1_Hadamard_14160_PaleyI`; `2026-10-08_Band1_Hadamard_14144_PaleyI`; `2026-10-08_Band1_Hadamard_14108_PaleyI`; `2026-10-08_Band1_Hadamard_14088_PaleyI`; `2026-10-08_Band1_Hadamard_14084_PaleyI`; `2026-10-08_Band1_Hadamard_14072_PaleyI`; `2026-10-08_Band1_Hadamard_14052_PaleyI`; `2026-10-08_Band1_Hadamard_14012_PaleyI`; `2026-10-07_Band1_Hadamard_14000_PaleyI`; `2026-10-07_Band1_Hadamard_13968_PaleyI`; `2026-10-07_Band1_Hadamard_13964_PaleyI`; `2026-10-07_Band1_Hadamard_13932_PaleyI`; `2026-10-07_Band1_Hadamard_13908_PaleyI`; `2026-10-07_Band1_Hadamard_13904_PaleyI`; `2026-10-07_Band1_Hadamard_13884_PaleyI`; `2026-10-07_Band1_Hadamard_13880_PaleyI`; `2026-10-07_Band1_Hadamard_13860_PaleyI`; `2026-10-07_Band1_Hadamard_13832_PaleyI`; `2026-10-07_Band1_Hadamard_13808_PaleyI`; `2026-10-07_Band1_Hadamard_13800_PaleyI`; `2026-10-07_Band1_Hadamard_13764_PaleyI`; `2026-10-07_Band1_Hadamard_13760_PaleyI`; `2026-10-07_Band1_Hadamard_13752_PaleyI`; `2026-10-07_Band1_Hadamard_13724_PaleyI`; `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`; `2026-10-06_Band1_Hadamard_13680_PaleyI`; `2026-10-06_Band1_Hadamard_13628_PaleyI`; `2026-10-06_Band1_Hadamard_13620_PaleyI`; `2026-10-06_Band1_Hadamard_13592_PaleyI`; `2026-10-06_Band1_Hadamard_13568_PaleyI`; `2026-10-06_Band1_Hadamard_13524_PaleyI`; `2026-10-06_Band1_Hadamard_13500_PaleyI`; `2026-10-06_Band1_Hadamard_13488_PaleyI`; `2026-10-06_Band1_Hadamard_13464_PaleyI`; `2026-10-06_Band1_Hadamard_13452_PaleyI`; `2026-10-06_Band1_Hadamard_13412_PaleyI`; `2026-10-06_Band1_Hadamard_13400_PaleyI`; `2026-10-06_Band1_Hadamard_13368_PaleyI`; `2026-10-06_Band1_Hadamard_13340_PaleyI`; `2026-10-06_Band1_Hadamard_13332_PaleyI`; `2026-10-06_Band1_Hadamard_13328_PaleyI`; `2026-10-06_Band1_Hadamard_13292_PaleyI`; `2026-10-06_Band1_Hadamard_13268_PaleyI`;

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2026-09-29_Band1_Hadamard_11172_PaleyI; 2026-09-29_Band1_Hadamard_11160_PaleyI;
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2026-09-24_Band1_Hadamard_10304_PaleyI; 2026-09-24_Band1_Hadamard_10272_PaleyI;
2026-09-23_Band1_Hadamard_10268_PaleyI; 2026-09-23_Band1_Hadamard_10224_PaleyI;
2026-09-23_Band1_Hadamard_10212_PaleyI; 2026-09-23_Band1_Hadamard_10164_PaleyI;
2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
2026-09-23_Band1_Hadamard_10140_PaleyI; 2026-09-23_Band1_Hadamard_10112_PaleyI;
2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_14304_paleyI/`.

Cutoff. 2026-10-09T11:35:14Z (~7:35AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14304 only (verified max|HHˆT-nI|=0), via classical Paley-I q=14303 (prime ===3 mod 4). 14304 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14304: Paley-I **PASS** ($q = 14303$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=7151 prime but ===3 mod 4 needs ===1); Kronecker **FAIL/SKIP** (note $14304 = 4 \cdot 3576$; $3576 \equiv 0 \pmod 4$ order-admissible but not_in_seeds / no H_3576; H_3576 has NO direct classical Paley route; $3576 = 2^3 \cdot 3 \cdot 149$; do NOT invent H_3576). **14304** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14252+14244+14208+14160+14144+14108+14088+14084+14072+14052+14012+14000+13968+139** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **14304**.

- H_{14304} = Paley-I ($q = 14303$) – classical SPK
- Paley-II for 14304: **FAIL/SKIP** ($q = 7151$ prime but $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 14304: **FAIL/SKIP** ($4 \cdot 3576$; $3576 \equiv 0 \pmod 4$ order-admissible but H_3576 has NO direct classical Paley route; $3576 = 2^3 \cdot 3 \cdot 149$)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14304.py`

Results. overall_pass=True; units 678/678; row blocks 112/112 (block size 128); wall 2767.5075s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-09T11:35:14Z (~7:35AM ET)

Arithmetic

- Target $n = 14304 = q + 1$ with $q = 14303$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 7151$ prime but $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $14304 = 4 \cdot 3576$; $3576 \equiv 0 \pmod 4$ order-admissible but not_in_seeds / no H_3576; H_3576 has NO direct classical Paley route; $3576 = 2^3 \cdot 3 \cdot 149$ — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14304 Paley-I	Paley-I q=14303	0	2725.7243
H_14304 Paley-II	FAIL/SKIP (q=7151 prime but ===3 mod 4 needs ===1)	–	–
H_14304 Kronecker	FAIL/SKIP (4*3576; 3576===0 mod 4 order-admissible; H_3576 has NO direct classical Paley route)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=14303)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^2*71317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^2*59^2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3^2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2^2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3^2*389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2^2*3^2*389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13^2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*1167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2^2*3*1167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=5*2941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2^2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3^2*1571 NOT prime -> FAIL; Kronecker 14140=4*3535; 3535=5*7101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2^2*5*7101 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=4*3555; 3555=3^2*5*79 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555);

14220=2^{2}*3^2579 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7129->n=14260 Paley-II DIRECT (skipp*
*NOT prime -> FAIL; Kronecker 14260=4*3565; 3565=5*23*31 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^2*5*23*31 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | next Paley-I DIRECT: q=14323->n=14324 (prime ===3 mod 4; no prime ===3 mod 4 in 14304..14322; 14321 ===1 skipped; Paley-II q2=7161=3*7*11*31 NOT prime power (===1 mod 4 but composite) -> FAIL/SKIP; Kronecker 14324=4*3581; 3581 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3581); 14324=2^2*3581 ===4 mod 16; FIRST any direct route after 14304) | Kronecker-only (no direct Paley route for n; note-only, NOT on box): 14320=4*3580 (H_3580 via Paley-II q2=1789 prime ===1 mod 4; 14320=2?*5179; Paley-I q=14319=3^2*3743 NOT prime; Paley-II q2=7159 prime but ===3 mod 4) – 14308=2^2*7^2*73, 14312=2^3*1789 (===8 mod 16), 14316=2^2*3*1193 have NO direct route and m=n/4 not Hadamard-admissible -> FAIL/SKIP | FIRST Paley-II DIRECT after 14304: q2=7177->n=14356 (q2=7177 prime ===1 mod 4; Paley-I q=14355=3^2*5*11*29 NOT prime -> FAIL; Kronecker 14356=4*3589; 3589=37*97 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3589); 14356=2^2*37*97 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); comes AFTER the next Paley-I 14324) (or gap 2052 / Track B)